

Technological ubiquity, social media and their effects on health: the importance of AI from robots to voice assistants

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Abstract

This essay explores the multifaceted impacts of technological ubiquity on individuals affected by autism and eating disorders, highlighting how digitalization shapes their experiences and healthcare interactions. Key drivers of technological ubiquity include rapid advancements in artificial intelligence, biotechnology, and increased accessibility to technology. The role of robotics, particularly in the form of interactive humanoid robots like NAO, illustrates positive applications in autism therapy by facilitating emotional recognition and interpersonal skills. The integration of voice assistants in healthcare settings exemplifies how technology can enhance patient support and streamline medical processes. However, the proliferation of digital media also poses risks for individuals with eating disorders, as social media can foster negative comparisons and unhealthy behaviors. The implications of these technological changes necessitate a thoughtful examination of ethical considerations, data privacy, and the potential biases inherent in AI systems. Ultimately, this essay emphasizes the importance of responsibly navigating the challenges and opportunities presented by technological ubiquity in healthcare.

Keywords

Technological Ubiquity, Digitalization, Healthcare, Autism, Eating Disorders, Artificial Intelligence, Humanoid Robots, Voice Assistants, Patient Support, Social Media, Ethical Considerations, Data Privacy, AI Biases.

Introduction

Technological ubiquity is particularly relevant to the healthcare system and the impact on people affected by autism and eating disorders, in addition to patients who need medical support: this essay will explore the ways in which digitalisation affects these individuals.

Drivers of technological ubiquity

One of the key drivers of technological ubiquity is the rapid pace of technological development and innovation. Advances in fields such as computer science, artificial intelligence, and biotechnology have led to the creation of new devices that have greatly expanded our capabilities and changed the way people live and work (Van Dijk, J., 2012). Another factor contributing to technological ubiquity is the continuously increasing affordability and accessibility of technology. As costs have fallen, more and more people have been able to afford and use advanced services, leading to a democratization of technology.

The internet and social media operate on the premise of constant connectivity, that is accompanied by a second major innovation: the mobility of personal devices in space, which are becoming more portable and, in some cases, wearable, and slowly becoming an extensions of the users' physical selves (Van Dijk, J., 2012).

The growing prevalence of robots and other forms of automation have been major contributors to the phenomenon of technological ubiquity (Bock, T., 2015). The rise of robotics has been facilitated by advances in fields such as artificial intelligence and machine learning. These technologies have made it possible for robots to perform complex tasks and make decisions based on data and information, greatly increasing their skills and versatility. Moreover, artificial intelligence-based systems and machine learning processes have gradually been incorporated into these voice assistants that synthesise human speech and are able to engage in conversations. However, voice assistants go beyond that. Research into their potential use-cases is being sparked by the idea of offering medical support to particular patients that could greatly benefit from such enhanced instruments (Sezgin, E. et al., 2020).

The Diagnostic and Statistical Manual of Mental Disorders defines autism a spectrum disorder because the way it manifests itself varies depending on the degree of the condition, a person's developmental stage, and their age (Widiger, T., 2011). It is characterized by challenges with social interaction, repetitive behaviours, and difficulty with verbal and nonverbal communication. People with autism often have difficulty interpreting social cues and expressing themselves, which can make it challenging for them to form and maintain relationships with others. They may also have unusual interests or behaviours that can be recurring or restrictive. Positive applications of technological ubiquity are the so-called kid-friendly robots with the goal of involving autistic children in certain games to help them imitate and respond appropriately to other people's emotions. This helps the child to improve autistic symptoms related to interpersonal deficits, while also implementing interventions that are adapted to their needs and young age (Riek, L.D., 2017).

NAO, developed by the French company Aldebaran Robotics, is a humanoid robot that can learn and tailor its responses based on the information it receives and discerns from the autistic child. Treatments for autism spectrum conditions must be specifically tailored to the weaknesses the kid exhibits while interacting with others or when their emotive and cognitive styles develop in an unusual way. Rudovic, O. et al. (2018) contend that machine learning and artificial intelligence systems can address the issues brought on by this particular disorder because they have the capacity to store and process a sizable amount of data that is as varied and heterogeneous as the needs of each autistic child. The goal of a therapy session is to teach the child to recognise emotions and then teach him how to replicate them, by showing them pictures of other children's faces in various emotional states. The humanoid robot enters the session and imitates the same emotions that the child has seen in the pictures. It then starts to interact with the child by occasionally changing the tone of its voice and moving its limbs (Feng, H. et al., 2013).

The research's main finding is that NAO can interact with kids in a way that makes it seem like a real person to them rather than just a toy. As a result, the information the humanoid gathers from the child's natural interaction is relatively trustworthy and spontaneous. Additionally, youngsters find interactions with robots way less frustrating than those with adults, who frequently adopt several, sophisticated facial expressions (Rudovic, O. et al., 2018). NAO's ability to learn from data directly provided by each child, including films of their facial expressions, bodily movements, gestures, and postures, audio files, and physiological measurements of their heart rates, body temperatures, and skin conductance responses, is what makes it so special (Rudovic, O. et al., 2018). Afterwards, researchers code the audio and video recordings as well as the child's psychophysiological data to determine how happy, distracted, interested, or engaged the autistic youngster was during the session. Following training on this data, NAO's deep learning network assists it in estimating the child's likely behaviour throughout the engagement (LeCun, Y. et al., 2015).

Connections to healthcare and implications

Connections between the usage of technology and the healthcare setting are many and take on real forms because of the possible applications of these voice assistants. An illustration of this is Boston Children's Hospital, which has implemented voice technology by conducting three trials: in the intensive care unit, the voice assistant supports professionals in organisational aspects, in the transplant unit it serves as a quick and direct interface for preliminary checks, and outside of the hospital it aids patients with typical and minor illnesses like fevers and colds (Wilder, J. et al., 2019). Furthermore, the National Health Service has partnered with Amazon to develop a suitable app which allows users to ask Alexa for medical advice and receive suggestions based solely on information that has been confirmed by the NHS (Brown, S., 2019). The primary beneficiaries of this programme are the elderly and the blind, demonstrating the healthcare provider's desire to enhance care for those who may most benefit from ongoing digital support in

the comfort of their own homes while reducing infrastructure stress to healthcare facilities (Dojchinovski, D. et al., 2019). With a technology capable of responding to the patient's demands and making autonomous decisions, the employment of the voice assistant as a tool for patient diagnosis and treatment aims to realise a true Ambient Assisted Living (Ben Hmida, H. and Braun, A., 2017). This will be accomplished by using the voice assistant as a tool for patient diagnosis and treatment. Natural language, which enables face-to-face communication and guarantees exceptional usability, helps to make these operations even more efficient.

The voice assistant may also be a crucial tool for chronic illnesses that are extremely severe and need for both homecare for the patient and communication with medical professionals to track symptoms and enable early treatment in case of acute crises (Moret-Tatay, C. et al., 2021). The ability of the virtual assistant to constantly interact with them and adapt to their requirements, even as they vary over time, could enable the development of support programmes for those responsible for the patient's care at home as well as for the patient himself. The currently available technologies for recording symptoms are primarily limited to the use of a screen and frequently need the user to go through a number of steps in order to record specific symptomatic events by choosing pre-set alternatives without having the ability to directly record symptoms (Kalem, G. and Turhan, Ç., 2015). The user may choose to omit some reports or make them less correctly due to the time and effort required to accurately record each occurrence, particularly for diseases with complicated symptomatology. In this situation, the use of natural language and the immediate engagement provided by the voice assistant may be able to remove these barriers and, if necessary, serve as a tool to acquire real-time cues on how to handle situations like acute crises or atypical symptoms.

Technologies have always supported and shaped therapeutic procedures. Tools using artificial intelligence directly communicate with patients through speech synthesis, replicating the customary oral form of communication, in

addition to modelling human behaviour in managing and solving complicated problems. These new tools establish entirely new patient-physician relationships that are soon to occasionally become fully autonomous (Bohr, A. and Memarzadeh, K., 2020). The limitation to keep in mind with even the most sophisticated voice assistants is that they will unavoidably have biases due to the kind of training received, the data utilised to teach the algorithms to reason, and the absence of empathy with regard to the conventional human interaction. This may result in rationally ideal diagnoses and treatments, but they may be ineffectual due to factors such as the individuality of each person, the sociocultural environment in which they are, generational gaps, cultural diversity that affects how people live their lives, among the others (Ermolina, A. and Tiberius, V., 2021). Therefore, it is essential to analyse these potential biases in order to be able to confirm or perhaps eliminate them.

On the other hand, the negative effects of technological ubiquity are related to a number of disorders that fall under the diagnostic category of eating and nutrition disorders. Social media have some features that make them particularly relevant to eating disorders, including a strong emphasis on the visual and photographic content (Holland, G. and Tiggemann, M., 2016). Moreover, through coded responses such as likes and comments, social media enable high degrees of continuous interaction. Social networking sites have the potential to tailor information and advertising content to a person, developing a more specialised online environment (Altendorf, M., 2011). It is possible to hypothesise that content targeting may cause advertisements for goods related to the worlds of fitness, diets, and cosmetics to specifically target people who have negative experiences with their bodies. Furthermore, social networks can create a sense of comparison and competition among users. Seeing others posting about their workouts, diets, and weight loss can lead individuals to feel inadequate or like they are not doing enough. This can create a negative feedback loop, where individuals engage in disordered eating behaviours in an attempt to meet these perceived standards (Dakanalis, A., 2017).

The effects of digitalization and technological ubiquity on the healthcare system can be regarded from a sociological perspective as a reflection of the larger social and cultural milieu. A movement towards a more individualistic and technologically oriented society has occurred at the same time as the development of digital technology (Lupton, D., 2014). It is critical to take into account the requirements and difficulties of patients in order to support them in the era of technological ubiquity, by assisting them in using technology effectively and offering training and support as well as customising digital tools to suit their individual needs. The implementation of technologies in healthcare necessitates a renewed focus on adherence to the principles governing the protection of personal data, focusing not only on general elements typical of processing by means of artificial intelligence systems but also on specific aspects peculiar to voice assistants, primarily linked to the governance and security of processing. The primary concerns raised by automated decisions made through the use of neural network systems and, subsequently, on issues related to the voice assistant's key features, such as governance and technical-organizational security, must be questioned in order to steer current technological scenarios towards a type of development that is compliant with the protection of personal data. (Hasal, M. et al., 2021).

Conclusion

All things considered, the impact of technological ubiquity on society and individuals is complex and multifaceted. As we continue to move into a future that is increasingly shaped by technology, it is important to consider its implications and to address the potential challenges and negative effects that it may bring. This will require a concerted effort by individuals, organizations, and policymakers to ensure that technology is used in a way that is responsible, ethical, and beneficial for all members of society.

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